



PRODUCT NAME

High-performance Dishwashing Liquid Formula

Description

manual dishwashing detergent for kitchen use

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	14.85%	Primary surfactant providing cleaning and foaming
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	4.95%	Secondary surfactant enhancing foam and mildness
3	Sodium Citrate	Sodium Citrate	1.98%	Builder to enhance cleaning performance and softening water
4	Sodium Carbonate	Sodium Carbonate	2.97%	pH adjuster and builder to enhance detergent performance
5	Propylene Glycol	Propylene Glycol	2.48%	Solvent and humectant for improved texture and stability
6	Fragrance (Limonene)	Limonene	0.5%	Fragrance for pleasant scent
7	Preservative (Benzyl Alcohol)	Benzyl Alcohol	0.5%	Preservative to prevent microbial growth
8	Water (Deionized)	Water	71.77%	Solvent for the formulation
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions